Priority Irons

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```
## find "blue" pipes and fill gaps in data

library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.1.4     ✔ readr     2.1.5
## ✔ forcats   1.0.0     ✔ stringr   1.5.1
## ✔ ggplot2   4.0.0     ✔ tibble    3.2.1
## ✔ lubridate 1.9.4     ✔ tidyr     1.3.1
## ✔ purrr     1.0.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors
```

```
library(splines)

pipes <- read_csv("Train_update3.csv")
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for detail
s,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## Rows: 43039 Columns: 12
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr  (3): Pipe ID, Material, Surface
## dbl  (7): GPS x1, GPS y1, GPS x2, GPS y2, Leaked, Age, Length
## date (2): Lay date, Currdate
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
  # every pipe, with a Leaked flag (1 = leaked 2019–2026)
leaks <- read_csv("train.csv")
```

```
## Rows: 14113 Columns: 4
## ── Column specification ────────────────────────────────────────────────────
## Delimiter: ","
## chr  (1): Pipe ID
## dbl  (1): Cost
## date (1): Date
## time (1): Time
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
  # one row per recorded leak, with its repair cost

network <- pipes %>%
  mutate(
    lay_year = year(as.Date(`Lay date`)),
    material_missing = is.na(Material),
    Material_m = replace_na(Material, "wrought iron"),
    # 8 NA pipes: the project says every iron pipe must be replaced,
      # so assume iron to be safe
    # all 8 are on water surface, where wrought iron is most common

    lay_year_i  = coalesce(lay_year,
                # 15 pipes
                median(lay_year[Material_m == "wrought iron"], na.rm = TRUE)),
    is_iron = str_detect(Material_m, "iron"),
    mid_x = (`GPS x1` + `GPS x2`) / 2,
    mid_y = (`GPS y1` + `GPS y2`) / 2
      # use later for clustering/classification
  )

# blue pipes = never leaked (Leaked == 0, so not already replaced by a repair)
# AND not polyurethane to begin with (those never needed replacing)
needs_replacement <- network %>%
  filter(Leaked == 0, Material_m != "polyurethane")

nrow(needs_replacement)
```

```
## [1] 27248
```

```
# should be 27,248, matches Daniel's notebook
```

```
## fitting 1. leak model (splines + log) and 2. cost model (gamma)


# Likelihood: did the pipe leak in 2019–2026?
# logistic regression (binomial) because the outcome is yes/no.
leak_fit <- glm(
  Leaked ~ Material_m + Surface + ns(lay_year_i, 4) + log(Length),
  family = binomial,
  data = filter(network, Material_m != "polyurethane")
)

# Where:
# Leaked: whether the pipe actually leaked in 2019–2026 (1 = yes, 0 = no); the model pre
dicts the probability of Leaked = 1
# Material_m: the pipe's original material, with the 8 missing values filled in as wroug
ht iron
# Surface: what the pipe runs under (grassland, farmland, swamp, water, road, or structu
re)
# lay_year_i: the year the pipe was laid, with the 15 missing years filled in using the
median for wrought iron
# ns(lay_year_i, 4): lay year turned into a flexible curve so leak chance can rise and f
all with age, uses spline with df = 4
# log(Length): the log of the pipe's length in metres, so extra length matters less for
long pipes

# cost data: one row per leak, joined to pipe features
sev <- leaks %>%
  inner_join(select(network, `Pipe ID`, Material_m, Surface, Length), by = "Pipe ID")
# head(sev)

# cost given a leak (gamma model)
cost_fit <- glm(
  Cost ~ Surface + Material_m + log(Length),
  family = Gamma(link = "log"),
  data = sev
)
 summary(cost_fit)
```

```
##
## Call:
## glm(formula = Cost ~ Surface + Material_m + log(Length), family = Gamma(link = "lo
g"),
##     data = sev)
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)             8.310752   0.125215  66.372  < 2e-16 ***
## Surfacegrassland        0.038068   0.068129   0.559    0.576
## Surfaceroad             3.647180   0.103147  35.359  < 2e-16 ***
## Surfacestructure        2.159373   0.087870  24.575  < 2e-16 ***
## Surfaceswamp            0.018008   0.104402   0.172    0.863
## Surfacewater            0.004521   0.067186   0.067    0.946
## Material_mcast iron    -0.086195   0.099857  -0.863    0.388
## Material_mcopper       -1.275097   0.125112 -10.192  < 2e-16 ***
## Material_mgray iron     0.077328   0.118707   0.651    0.515
## Material_mwrought iron  0.033942   0.092937   0.365    0.715
## log(Length)             0.147033   0.031971   4.599 4.28e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 6.074767)
##
##     Null deviance: 50884  on 14112  degrees of freedom
## Residual deviance: 22946  on 14102  degrees of freedom
## AIC: 285497
##
## Number of Fisher Scoring iterations: 8
```

```
## creating a priority dataset based on expected cost if not fixed

# expected loss = P(leak) * cost if leak
priority <- needs_replacement %>%
  mutate(
    p_leak_8yr = predict(leak_fit, newdata = pick(everything()), type = "response"),
      # probability the pipe leaked over the 8-year window (2019-2026)

    p_leak_annual = 1 - (1 - p_leak_8yr)^(1/8),
      # geometric dist: Y ~ geom(p)

    cost_if_leak = predict(cost_fit, newdata = pick(everything()), type = "response"),
    exp_annual_loss = p_leak_annual * cost_if_leak,
        # expected yearly cost if left unreplaced

    loss_per_metre  = exp_annual_loss / Length
  )
head(priority)
```

```
## # A tibble: 6 × 24
##   `Pipe ID` `Lay date` Material  Surface   `GPS x1` `GPS y1` `GPS x2` `GPS y2`
##   <chr>     <date>     <chr>     <chr>        <dbl>    <dbl>    <dbl>    <dbl>
## 1 P01001    1955-03-11 copper    grassland    6136.    5021.    6124.    5021.
## 2 P01004    1982-05-22 cast iron structure    2837.    3378.    2849.    3405.
## 3 P01006    1971-10-05 cast iron structure    3740.    5080.    3728.    5120.
## 4 P01007    1971-10-06 cast iron structure    3734.    5099.    3722.    5099.
## 5 P01008    1971-10-05 cast iron structure    3734.    5099.    3728.    5077.
## 6 P01009    1982-05-29 cast iron water        2570.    4299.    2615.    4299.
## # ℹ 16 more variables: Leaked <dbl>, Currdate <date>, Age <dbl>, Length <dbl>,
## #   lay_year <dbl>, material_missing <lgl>, Material_m <chr>, lay_year_i <dbl>,
## #   is_iron <lgl>, mid_x <dbl>, mid_y <dbl>, p_leak_8yr <dbl>,
## #   p_leak_annual <dbl>, cost_if_leak <dbl>, exp_annual_loss <dbl>,
## #   loss_per_metre <dbl>
```

```
# write_csv(priority, "priority_all.csv")
```

```
## priority datasets (by iron)

# only iron pipes are mandated!
priority_iron <- priority %>%
  filter(is_iron)

# rank by expected loss per meter
iron_rank <- priority_iron %>%
  group_by(Material_m) |>
  summarise(loss_per_m = sum(exp_annual_loss) / sum(Length)) |>
  mutate(type_rank = min_rank(desc(loss_per_m)))
# 1 = worst alloy

# sort by expected annual loss
priority_iron <- priority_iron %>%
  left_join(select(iron_rank, Material_m, type_rank), by = "Material_m") %>%
  arrange(desc(exp_annual_loss)) %>%
  mutate(priority_rank = row_number()) %>%
  select(priority_rank,
         `Pipe ID`,
         iron_type = Material_m,
         type_rank,
         Surface,
         Length,
         p_leak_annual,
         cost_if_leak,
         exp_annual_loss,
         loss_per_metre,
         mid_x,
         mid_y,
         material_missing)

head(priority_iron)
```

```
## # A tibble: 6 × 13
##   priority_rank `Pipe ID` iron_type type_rank Surface Length p_leak_annual
##           <int> <chr>     <chr>         <int> <chr>    <dbl>         <dbl>
## 1             1 P06643    cast iron         3 road      73.1         0.196
## 2             2 P09247    cast iron         3 road      69.8         0.197
## 3             3 P09249    cast iron         3 road      69.8         0.197
## 4             4 P06641    cast iron         3 road      65.0         0.197
## 5             5 P06513    cast iron         3 road      62.6         0.197
## 6             6 P07444    cast iron         3 road      57.6         0.197
## # ℹ 6 more variables: cost_if_leak <dbl>, exp_annual_loss <dbl>,
## #   loss_per_metre <dbl>, mid_x <dbl>, mid_y <dbl>, material_missing <lgl>
```

```
# write_csv(priority_iron, "priority_iron.csv")
```

```
## checking based on loss (total expected yearly loss, sorted by surface),
  # sanity check?
priority_iron %>%
  group_by(Surface) %>%
  summarise(n = n(), avg_loss = mean(exp_annual_loss), total_loss = sum(exp_annual_los
s)) %>%
  arrange(desc(total_loss))
```

```
## # A tibble: 6 × 4
##   Surface       n avg_loss total_loss
##   <chr>     <int>    <dbl>      <dbl>
## 1 road       1414    9616.  13597459.
## 2 structure  3409    1488.   5074083.
## 3 water      6909     262.   1810128.
## 4 grassland  8930     176.   1575952.
## 5 farmland   4235     216.    916305.
## 6 swamp      1348     256.    345685.
```

```
## just checking our datasets are lengthwise correct!

priority_all <- read_csv("priority_all.csv")
```

```
## Rows: 27248 Columns: 24
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr   (4): Pipe ID, Material, Surface, Material_m
## dbl  (16): GPS x1, GPS y1, GPS x2, GPS y2, Leaked, Age, Length, lay_year, la...
## lgl   (2): material_missing, is_iron
## date  (2): Lay date, Currdate
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
priority_iron <- read_csv("priority_iron.csv")
```

```
## Rows: 26245 Columns: 13
## ── Column specification ────────────────────────────────────────────────────
## Delimiter: ","
## chr (3): Pipe ID, iron_type, Surface
## dbl (9): priority_rank, type_rank, Length, p_leak_annual, cost_if_leak, exp_...
## lgl (1): material_missing
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
nrow(priority_all)
```

```
## [1] 27248
```

```
nrow(priority_iron)
```

```
## [1] 26245
```

```
nrow(priority)
```

```
## [1] 27248
```

```
# it's good! priority has copper + brass,
# while the new two take them out, so less rowss
```